

*Distributed systems expert with focus on
performance optimization*

Experience

- Sept. 2019–present **Senior Software Engineer I, Technical Lead, Senior Software Engineer II, HubSpot, Inc**, Cambridge, MA
- Maintained HubSpot's HBase installations, holding most of the company's data
 - Contributed improvements upstream to HBase project, earning committer status
 - Expanded HBase footprint from one to five AWS regions
 - Did whatever was necessary to fix problems: dove into HBase Java code, JVM internals, and the Linux kernel
 - Significantly improved HBase's performance for HubSpot's workloads by reducing garbage collection churn and byte array copies.
- June 2015–Jan. 2019 **Software Engineer I & II, Site Reliability Engineer II, Indeed, Inc**, Austin, TX
- Led spin-up of new data center
 - Rolled out system to incrementally load and validate data files to prevent bad artifacts causing full outages
 - Drove HTTP 1.1 to HTTP 2 switchover, improving page load speed 10%
 - Pioneered pattern for rewriting applications so they could run on Indeed's internal cloud, and personally refactored 10+ applications this way
 - Subject matter expert in Java and Spring within the SRE department, bringing my knowledge from the software role
 - Scaled up email sending pipeline from 50,000 to 50,000,000 emails per day

Publications & Affiliations

- Publications (linked) **It's Never a Hardware Bug, Until it is** (Discovery of how a CPU erratum affects the Z garbage collector in OpenJDK 21+)
- HubSpot Incident Report** (How a Linux kernel bug affected HBase servers)
- Isolating a Bug in the Java Virtual Machine** (Dive into an OpenJDK bug that I discovered)
- Affiliations **Committer on the Apache HBase project**

Education

- 2009–2014 **Bachelor of Science, Computer Science, Northeastern University**, Boston, MA
magna cum laude